

Experiment study on vortex-induced vibration of variable cross-section bridge tower considering alongwind and acrosswind coupling effect

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ABSTRACT

This study investigates the coupled responses between the alongwind and acrosswind directions of a variable cross-section bridge tower model through wind tunnel experiments. Uniform and two turbulent flows with four different wind directions are designed to study their influences. The results show a significant coupling effect between the alongwind and acrosswind responses, influenced by incoming flow conditions and structure damping. In uniform flow, two distinct vortex-induced vibration (VIV) regions are observed when the structural damping is low, possibly due to the variable cross-section. When coupled VIV occurs, the responses of both alongwind and acrosswind directions show a hardening non-Gaussian distribution, and the kurtosis value is close to 1.5. The increase of structural damping will weaken the coupling effect, but slightly increase the dominant frequency of the coupling. Turbulence intensity reduces the VIV effect and coupling effect but does not eliminate the coupling in galloping. In particular, the critical wind speed of galloping will decrease with the increase of turbulence intensity. The coupling effect is prominent at 0° wind direction, mainly dominated by the acrosswind direction. However, the coupling effect is weak in other wind directions and is primarily dominated by the alongwind direction. The coupling effect makes the energy transfer between the alongwind and acrosswind directions, which is crucial for designing variable cross-section high-rise buildings and tower structures susceptible to wind-induced vibration.

1. Introduction

High-rise buildings and towers are susceptible to wind-induced vibration under the flow-structure interaction due to their high flexibility and low damping characteristics, which have been studied in the literature [1–5]. When the vortex shedding frequency is close to the natural frequency of the structure, the phenomenon of “frequency lock-in” is prone to occur, resulting in a significant amplification of acrosswind vibration [6–9]. The mechanism of vortex-induced vibration (VIV) under atmospheric boundary layer wind flows is usually derived from the combined action of vortex shedding and incoming turbulence. VIV not only affects the comfort of the building but also increases the risk of fatigue damage to the structure.

Research on VIV mainly focuses on the acrosswind response because it is always the significant component in the design of high-rise flexible

structures [10–14]. The excitation mechanism of the acrosswind response is vortex shedding, making the structural vibration more challenging to predict than the alongwind response [2,15,16]. Close to the lock-in wind speed region, acrosswind response is predominantly affected by nonlinear negative aerodynamic damping, leading to a characteristic hardening behavior with a non-Gaussian distribution [7,9,17]. With the increase of wind speed, the acrosswind response is primarily driven by stochastic aerodynamic forces and follows a Gaussian distribution [7,17,18]. Alongwind response is usually neglected in VIV studies, because its excitation mechanism is turbulence buffeting, and response is much smaller than that in acrosswind direction, especially in lock-in wind speed region [15,19,20]. However, the mass eccentricity can alter the modal frequency and induce significant alongwind and acrosswind mode coupling [21,22]. The eccentricity influences not only the vibration magnitudes but also the correlations, subsequently affects

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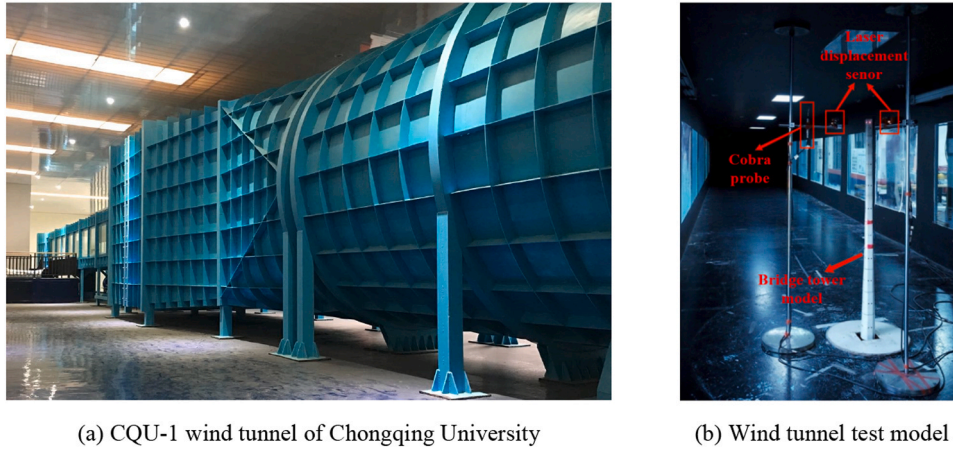


Fig. 1. Wind tunnel experiment.

the statistics of responses. The coupling effect will decrease the acrosswind response but increase the alongwind response, especially when the frequency ratio of alongwind and acrosswind direction is close to 1 [23–25]. The amplification of the alongwind vibration indicates the energy transfer from the acrosswind direction to the alongwind direction [24,26]. The aeroelastic model experiment in the wind tunnel effectively investigates VIV response with coupling effect. Some experiments show that acrosswind VIV may enhance the response of alongwind vibration in a certain range of wind speed, indicating the aerodynamic coupling effect [15,19,27,28]. Therefore, for the safety design of the flexible structure, it is essential to consider the coupling effect.

The coupling effect of alongwind and acrosswind vibration is prominent for square high-rise buildings [15,19,27]. Li et al. [15] compared the SDOF (Degree of Freedom) and 2DOF model responses and found that the coupling vibrations of building structures are stronger when the stiffness ratio is close to 1. Through the 2DOF rocking model wind tunnel test under low and high turbulence conditions, Yang et al. [29] showed that the cylindrical structure also had coupling effects. However, when the shape of the structure changes, the VIV characteristics are different from those of traditional square high-rise and cylindrical tower structures [30,31]. In the fluid, for the variable cross-section cylinder, the gradual change of the cross-section shape will affect the distribution of the fluid on the surface of the structure, and the vortex around it shows chaotic characteristics. Especially in the position where the cross section changes obviously, the shape of the vortex becomes more irregular [32–34]. In the wind field, the research on VIV of variable cross-section structures is not sufficient, and the gradual change

of cross-section shape may also affect its VIV characteristics. Study on the VIV and coupling effect for variable cross-section structure need to be further explored.

In this paper, the VIV response and influence of coupling effect on dynamic responses of the variable cross-section bridge tower model under different wind field conditions are studied through wind tunnel experiments. The characteristics of VIV response in both directions are comprehensively investigated. The amplitude spectrum and response correlation are then analyzed to further demonstrate the coupling effect. The non-Gaussian characteristic is obtained through Kurtosis and probability density function. The main structure of this paper is as follows: Section 2 introduces the wind tunnel experiment and the design of the test model. In Section 3, the vortex-vibration coupling effect under different flow field conditions is analyzed. Section 4 illustrates the influence of incoming wind directions on the coupling effect. Section 5 summarizes the main findings of this study.

2. Wind tunnel test on the wind-induced response of a bridge tower

2.1. Simulated approaching flow

The wind tunnel experiment was conducted in the direct-flow wind tunnel CQU-1 at Chongqing University, China. The size of the test section is 2.4 m (width) $\times$ 1.8 m (height) $\times$ 15 m (length), and the test wind speed range is 0.5 $\sim$ 35 m/s. The wind tunnel is shown in Fig. 1(a). Two types of flow are considered in this study: uniform flow and turbulent flow. The turbulent flow field with different turbulence

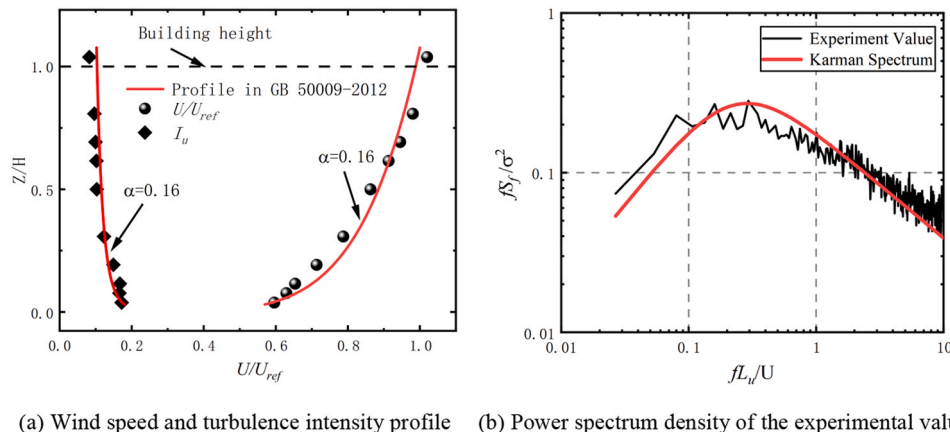


Fig. 2. Simulated wind flow in the wind tunnel.

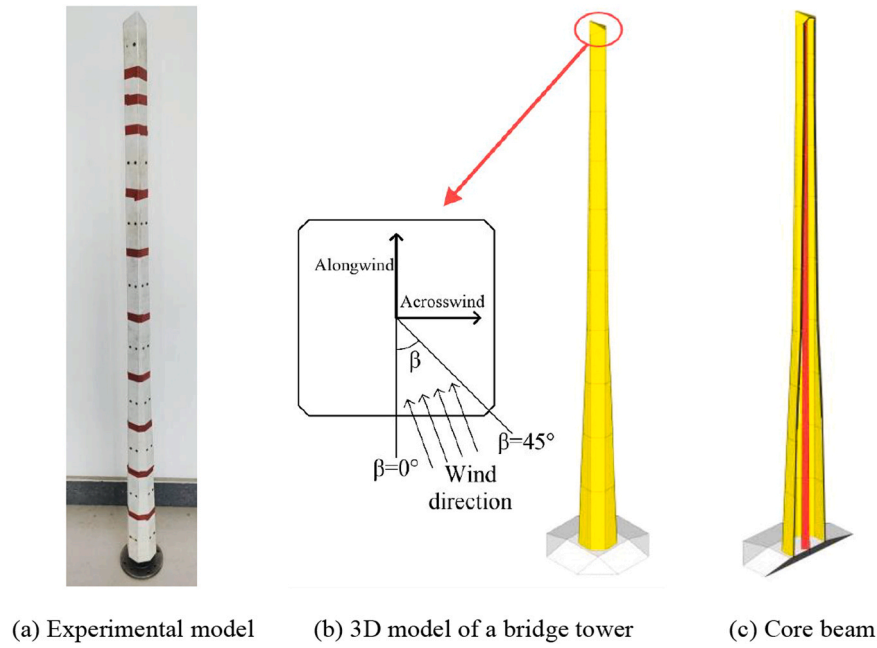


Fig. 3. Design of the experimental model.

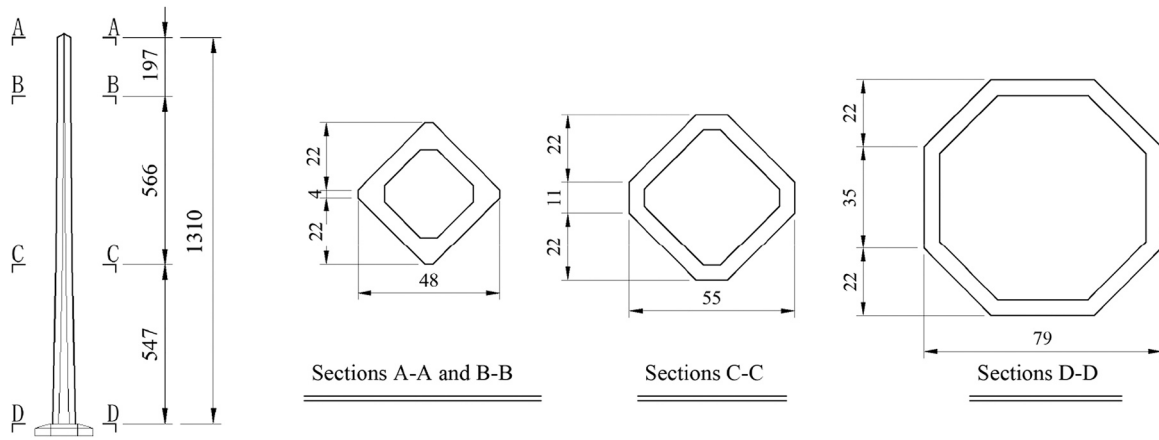


Fig. 4. Details of the aeroelastic model (unit: mm).

intensities is simulated by setting two different sizes of grilles at the entrance of the wind tunnel test section. The wind speed is measured by a cobra anemometer, and the turbulence parameters of the three types of flow fields are determined. The response is measured by installing a laser displacement meter on both sides of the top of the bridge tower, as shown in Fig. 1(b).

The incident wind in open terrain was simulated in accordance with code GB50009-2012, as shown in Fig. 2(a). The turbulence intensity (I_u) at the top of the model is 8.5 %, and the power law index (α) is 0.16. The

power spectrum density of alongwind turbulence is in good agreement with the Von Karman spectrum, as shown in Fig. 2(b). Reference height L_u is the height of tower top.

2.2. Aeroelastic model and its parameters

The variable cross-section tower of a cable-stayed bridge is used in this study. The height from the foundation to the top of the tower is about 157 m. It is a single-column concrete tower characterized by a square cross-section at the top with a straight 45° chamfer on each corner. The upper section maintains a constant cross-section, while the width of the chamfer faces in the lower and middle sections varies linearly with tower height. From the bottom to the top of the bridge tower, there is a degradation trend from octagon to chamfered quadrilateral.

According to the test section of the wind tunnel at Chongqing University, China, a 1:120 scale model of a bridge tower was fabricated to adhere to the non-dimensional similarity criteria. The model stands at a height of 1.31 m with a top width of 0.034 m, meeting the requirement of wind tunnel experiments for a maximum blockage ratio of less than 5 %. The schematic diagram of the bridge tower model used in this

Table 1
Parameters of the bridge tower model.

Terms	Aeroelastic model	
Similarity ratio	1:120	
Height (m)	1.31	
Top width B (cm)	3.4	
Frequency (Hz)	Alongwind	3.96
	Acrosswind	3.80
Damping ratio	Initial	0.03 %
	Tested	0.09 %–1 %
Wing direction	0° – 45°	

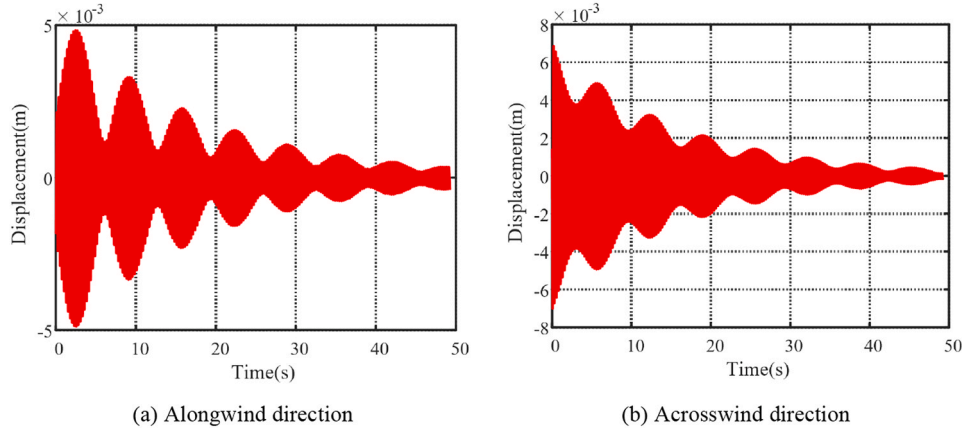


Fig. 5. Time histories of free vibration decay.

experiment is shown in Fig. 3(a). Fig. 3(b) shows the three-dimensional modeling shape of the bridge tower and the definition of the flow direction. The metal core beam provides the stiffness of the bridge tower, and the aerodynamic shape is provided by the 3D-printed outer membrane. These two components are fixed together by a support bracket. The completed core beam and the fully assembled bridge tower are shown in Fig. 3(c). The shape and size of the bridge tower and the cross-sectional dimensions of different heights are shown in Fig. 4. Table 1 lists the parameter information of the bridge tower model. The frequencies of the bridge tower in the acrosswind and alongwind directions are 3.80 Hz and 3.96 Hz, respectively, which are nearly 1:1.

The structure model has ten membrane segments, each with a 2 mm gap reserved between them to prevent collisions and avoid affecting structural damping. Two laser displacement sensors and a pitot tube anemometer are mounted on the support structure to record the displacement-time data of the structure response and the wind speed-time data, respectively. The structural damping ratio of the test model is identified using the free vibration decay method with the time histories shown in Fig. 5, indicating an initial damping ratio of 0.03 %. In the experiment, it is feasible to conveniently increase the structural damping ratio by pasting tape in the membrane gaps [35]. Five damping ratios ranging from 0.03 % to 1.0 % were utilized in this study. For each structural damping level, the reduced wind speed of the test was set from 4 to 23, and the sampling frequency of the wind speed and displacement data was 200 Hz.

3. Vortex-induced vibration characteristics of the coupling effect

3.1. Response results from wind tunnel tests

The 400 s dynamic displacements of both acrosswind and alongwind directions are simultaneously measured. The first 250 s displacement result is removed to eliminate the effect of transient structural dynamics. Fig. 6 shows the normalized acrosswind displacement y/B and alongwind displacement x/B under three different reduced wind speeds at uniform flow when the structural damping is 0.09% and wind direction is 0° . It is obvious that the acrosswind displacement results show more regular variation during the recorded time when $U/f_s B = 8.54$ and 18.51. This acrosswind VIV is dominated by self-excited forces and is investigated by previous scholars [1,17,36]. The response in alongwind direction, which should be the stochastic response, also shows steady sinusoidal vibration. The steady response in alongwind direction should be induced by the acrosswind response and indicate the coupling effect, which should be caused by the eccentricity. The response in acrosswind direction is larger than that in alongwind direction. When $U/f_s B = 12.77$, the measured responses in two directions are close and have

more stochastic features. With only wind flow in alongwind direction, the acrosswind vibration should be induced from the eccentricity of mass [21,22].

Fig. 7 shows the standard deviations (STDs) of displacement response in two directions under uniform flow. When $\zeta_s = 0.09\%$, the displacement STDs show different characteristics at reduced wind speeds. It is clear that there are two wind speed regions (i.e., $U/f_s B \approx 9$ and 18) for vortex-induced resonance, indicated as “VIV-I” and “VIV-II” respectively. In the aeroelastic experiment of the square high-rise model, there is only one VIV region observed and similar to “VIV-II” [29,37]. The occurrence of “VIV-I” maybe because the cross-section is variable. Generally, the STD of alongwind response will increase monotonically with increased wind speed. However, Fig. 7(a) indicates that the alongwind and acrosswind responses change synchronously. The STDs firstly are small and then rapidly increase after $U/f_s B = 8.1$ at the vicinity of VIV-I. Then, STDs become small at $U/f_s B = 12.7$, then rapidly increase again at the vicinity of the VIV-II region and become small at $U/f_s B = 20$. The results suggest that the vortex shedding of the acrosswind direction affects the flow field of alongwind direction, and the coupling effect with close 1:1 frequency ratio will cause a significant increase in alongwind response.

The STDs at other structural damping, i.e., $\zeta_s = 0.3\%$, 0.5% and 1%, are also shown in Fig. 7. The alongwind response exhibits a similar trend to the acrosswind response at different damping ratios. The responses near VIV-I and VIV-II are similar to those in acrosswind direction, albeit with amplitudes only one-tenth of those in the acrosswind response. The response STDs decrease with increasing structural damping ratio, which indicates the nonlinearity of aerodynamic damping. The response STD of VIV-II is smaller than that of VIV-I at lower structural damping ($\zeta_s < 0.5\%$), but is close to higher structural damping ratios ($\zeta_s \geq 0.5\%$). It is noted that the STDs increase dramatically after $U/f_s B = 26.9$ when $\zeta_s = 0.09\%$, and the displacement will exceed the measuring range of the displacement sensor. With the increased structural damping, the STDs sharply decrease. This phenomenon can be considered as galloping without nonlinear characteristics.

Fig. 8 shows the comparison of STDs of alongwind and acrosswind direction among uniform, low turbulence intensity ($I_u = 6.3\%$) and high turbulence intensity ($I_u = 13.1\%$) flow with lower structural damping ratios. The results in acrosswind direction show that the high turbulence intensity flow significantly affects the structural response and suppresses the VIV. Under high turbulence intensity flow, the structural response increases almost monotonically after $U/f_s B = 8.54$, and there is no obvious VIV. The response is dominated by buffeting forces and shows clear stochastic features.

The low turbulence intensity flow will also decrease the STD, whereas the response is still dominated by self-excited force. The characteristic in the alongwind direction is similar. The coupling effect is

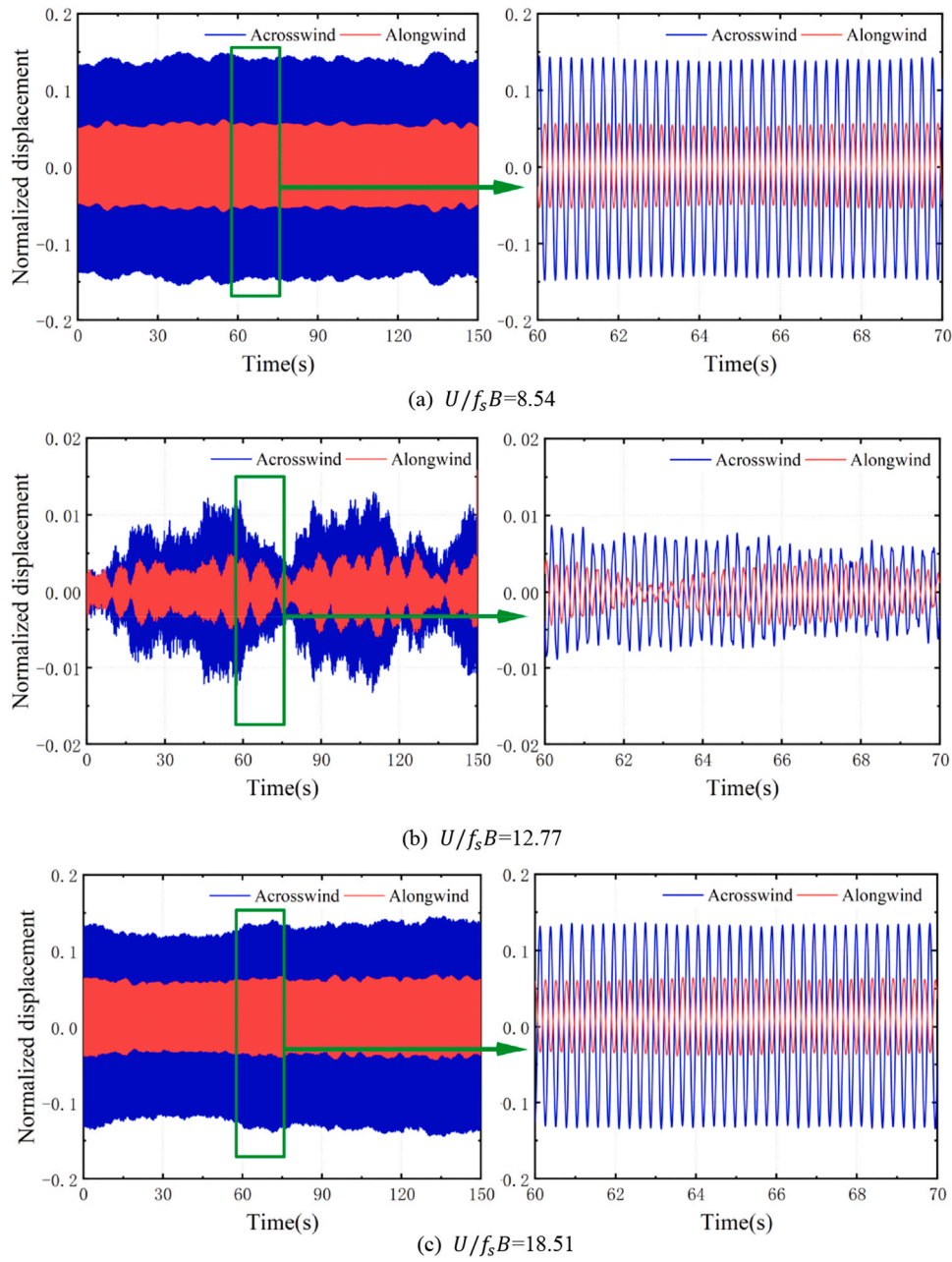


Fig. 6. Measured displacements in two directions at uniform flow ($\zeta_s = 0.09\%$).

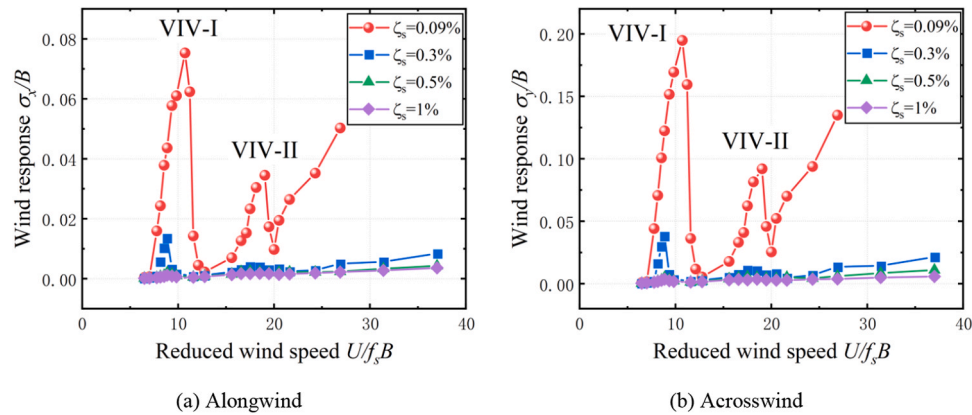


Fig. 7. STDs of displacement response of uniform flow.

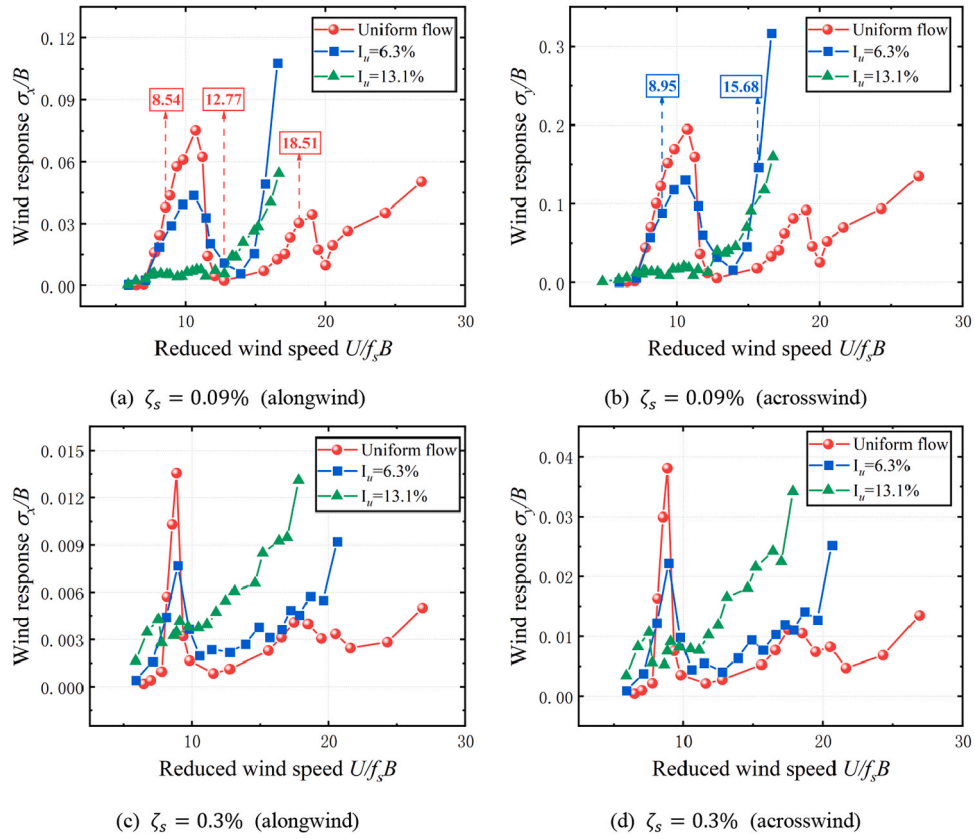
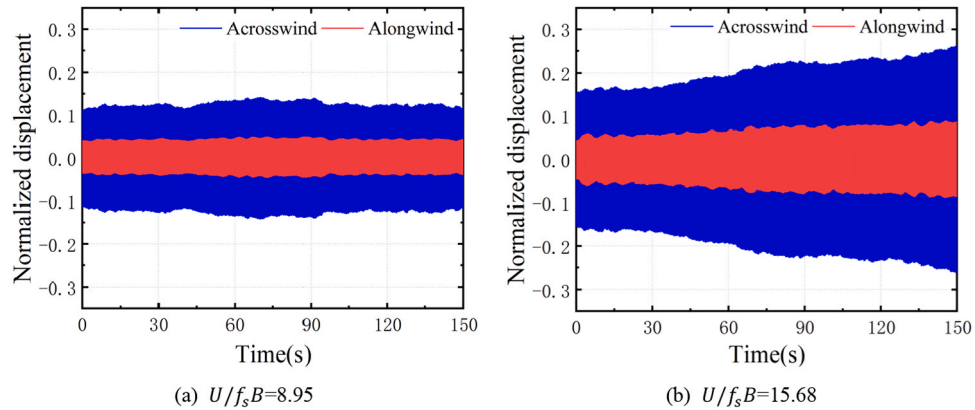


Fig. 8. STDs under different flow fields.

Fig. 9. Displacement response of turbulent flow under different reduced wind speeds ($I_u = 6.3\%$).

also significant in low turbulence intensity flow. However, the VIV-II region, which is apparent in uniform flow, is not observed in turbulent flow. The difference in turbulence effect on the response of VIV-I and VIV-II indicates that the mechanism of these two regions should be different. The turbulence intensity will decrease the critical wind speed of galloping. For example, it can be seen from Fig. 8(b) that the critical wind speed of galloping is 20 in uniform flow, 14 in low turbulence intensity ($I_u = 6.3\%$) and 12 in high turbulence intensity ($I_u = 13.1\%$). The decrease of critical wind speed with the increased turbulence intensity can be explained by the interaction of VIV and galloping [38,39].

Fig. 9 shows the displacement time history of when $I_u = 6.3$. Fig. 9(a) shows the displacement time history in VIV-I. The vibration response is a stable harmonic process, but smaller than the corresponding results in uniform flow. Fig. 9(b) shows the displacement time history at $U/f_s B = 15.68$. When the wind speed is located at VIV-II of the uniform flow,

stable response is not observed at turbulence flow, and the displacement response shows a gradually increasing galloping phenomenon.

3.2. Amplitude spectra

Fig. 10 shows the dominant frequency at different reduced wind speeds under different structural damping in uniform and turbulent flows. The natural frequencies of alongwind and acrosswind are 3.80 Hz and 3.96 Hz, respectively. From results with low structural damping, it can be found that the dominant vibration frequencies of these two directions in both VIV and non-VIV regions are almost the same and close to the natural frequency of acrosswind direction, except when wind speed is low. With the increased structural damping ratio, the dominant frequencies in two directions will be more likely to be their frequencies. The difference in dominant frequencies represents the decrease in

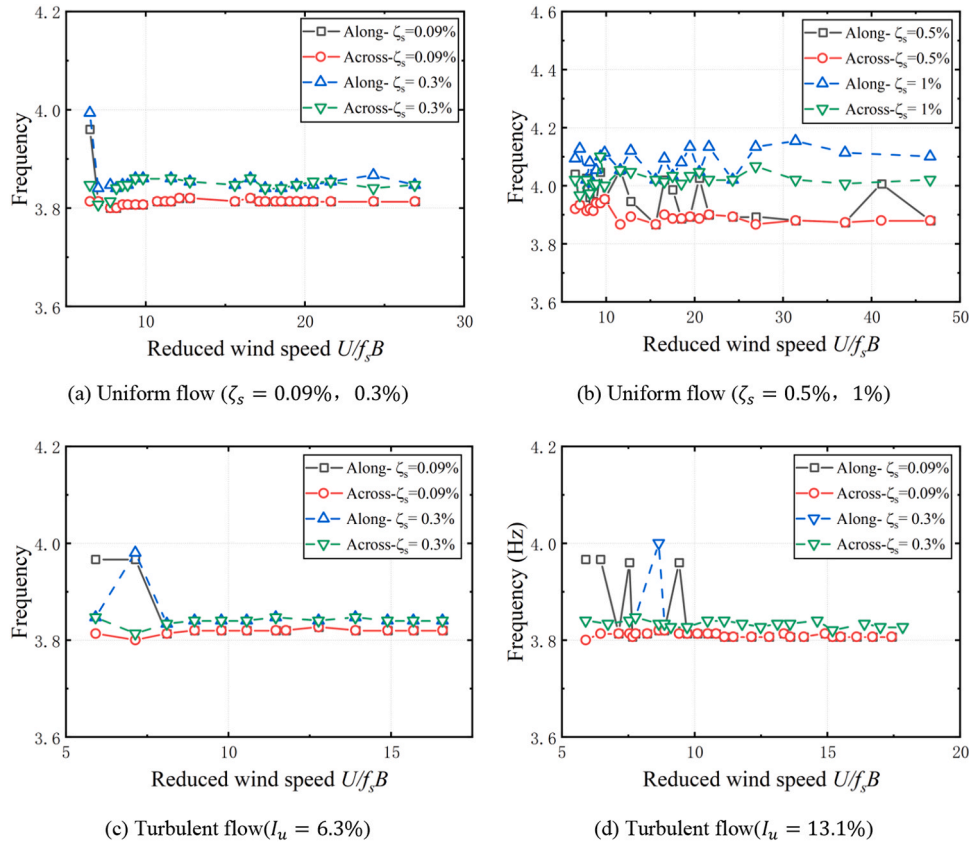


Fig. 10. The dominant frequency of response under different structural damping.

coupling degree. The dominant frequency of coupled vibration slightly increased with the structural damping ratio, possibly due to the effect of aerodynamic stiffness. Fig. 10 (c) and (d) are the dominant frequencies in turbulent flow. At the same structural damping, the coupling dominant frequencies of low turbulence intensity ($I_u = 6.3\%$) and high turbulence intensity ($I_u = 13.1\%$) are almost the same as those in uniform flow.

Power spectral density (PSD) of alongwind and acrosswind direction is further investigated to study the coupling effect (as shown in Fig. 11). The amplitude spectrums of four cases, i.e., before VIV-I $U/f_s B = 6.48$, at VIV-I $U/f_s B = 8.54$, between VIV-I and VIV-II $U/f_s B = 12.77$, and at VIV-II $U/f_s B = 18.51$, at uniform flow when $\zeta_s = 0.09\%$ are shown here. At $U/f_s B = 6.48$, the coupled frequency is observed, but the coupling effect is weak. The dominant frequency in alongwind direction is its own natural frequency. At $U/f_s B = 12.77$, the coupling effect is also weak. However, the energy from the coupled frequency is larger than its natural frequency. Thus, the dominant frequency is the coupled frequency. As the frequency ratio is close to 1, coupled vibration frequencies will always occur when the wind blows over the model. However, it is crucial to emphasize that coupled frequencies do not necessarily indicate the occurrence of coupled vibration. In the two cases just discussed, as shown in Fig. 11 (a) and (c), more energy is concentrated at lower frequencies, and the response exhibits significant random vibration characteristics.

At $U/f_s B = 8.54$ and 18.51 , the dominant frequency in the alongwind direction is only the coupled frequency. The coupling effect is strong, and the contribution of coupled frequency to the response is significant. The energy from the alongwind natural frequency is weakened. In these cases, the vibration can be considered as coupled VIV with steady-state harmonic responses in both alongwind and acrosswind directions.

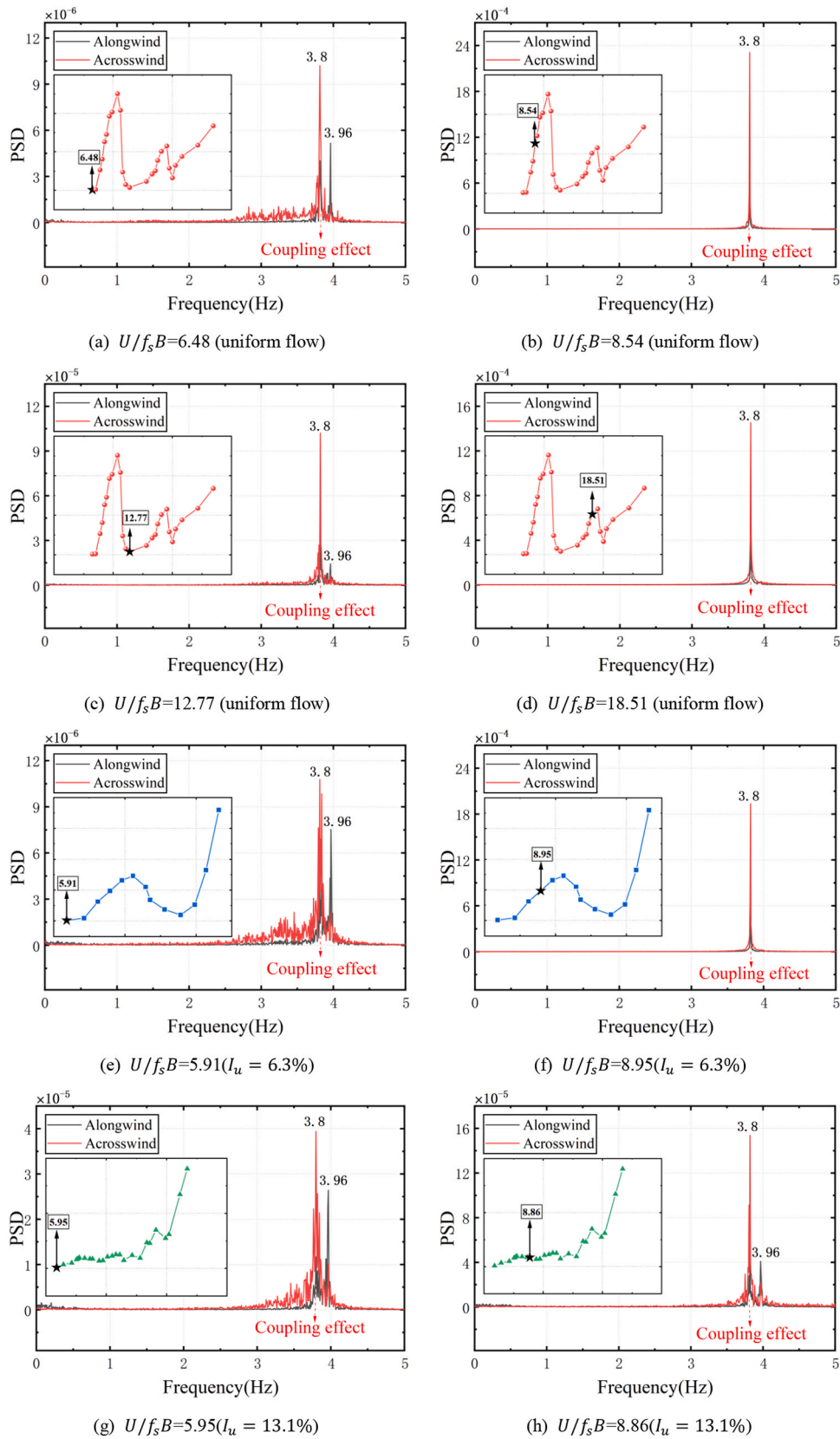
The amplitude spectra of turbulent flow are shown in Fig. 11 (e-h). At similar wind speeds, as the turbulence intensity increases, the low-

frequency energy increases accordingly, resulting in a displacement response closer to random vibration. When the coupling occurs under different wind speeds and different turbulence fields, the acrosswind vibration drives the alongwind vibration regardless the coupling effect is strong or weak. It shows that the coupling effect is dominated by the acrosswind direction.

3.3. Kurtosis and probability density function

The kurtosis value is exactly 3 for random processes, while for purely harmonic processes, it is 1.5. To further analyze the characteristics and coupling effects of alongwind and acrosswind responses statistically, Fig. 12 illustrates the kurtosis of the alongwind and acrosswind responses in uniform and turbulent flows. For uniform flow, kurtosis in two directions is similar. The kurtosis with high structural damping ratios fluctuates around 3, and indicates the response is a random process. For $\zeta_s = 0.09\%$, the kurtosis quickly drops from 3 to 1.5 when the wind speed is near the VIV region. However, for $\zeta_s = 0.3\%$, the kurtosis slightly decreases to around 2.5 in VIV-II, with no significant changes, indicating mixed random-harmonic responses. The decrease of kurtosis means an increase in the contribution of harmonic vibration. When $\zeta_s = 0.09\%$ and $U/f_s B = 21$, the kurtosis is close to 1.5, indicating the occurrence of galloping.

Compared with the results under uniform flow, the kurtosis under turbulent flow exhibits distinct characteristics. The contributions of harmonic motion decrease with the increase in turbulence intensity. However, the critical wind speed of galloping will decrease for both low and high turbulence intensity flow. The results further prove that turbulence will not suppress the coupling effect during galloping, which is different from its impact on VIV. When the wind speed is located in the VIV region, the response will decrease with the increase of turbulence intensity due to the suppression of VIV by turbulence. However, galloping is still observed with turbulence intensity after the VIV region.

Fig. 11. PSD under different flow fields ($\zeta_s = 0.09\%$).

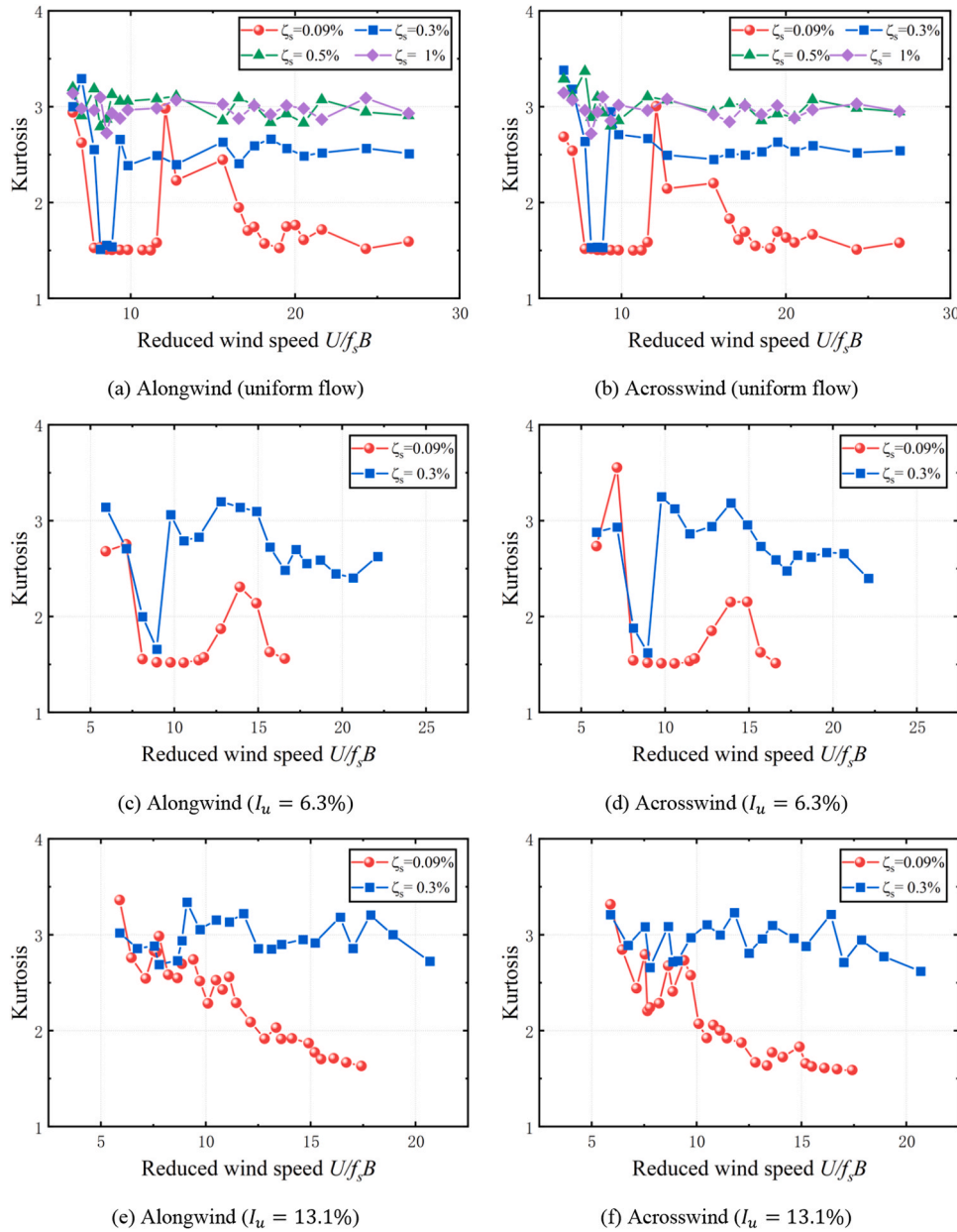


Fig. 12. Kurtosis of response under different flow fields.

In higher turbulence intensity fields, the coupling effect of the structure is only evident in the high wind speed galloping range for low structural damping.

Fig. 13(a-b) presents the probability density functions (PDFs) of the displacements under uniform flow. Before VIV-I, the response follows a Gaussian distribution. Between VIV-I and VIV-II, it follows a Gaussian-like distribution. The response is a combination of random vibration and harmonic vibration. In VIV-I and VIV-II, it follows a hardening non-Gaussian distribution, where the harmonic component dominates the response. The PDFs of the alongwind response exhibit the same trend as the acrosswind response. Fig. 13(c-d) presents the PDFs under low turbulent flow. In VIV and non-VIV regions, the PDFs of the displacements are similar to uniform flow, and the two directions are the same.

Fig. 14 shows the PDFs of the displacement response under different structural damping at the same reduced wind speed in VIV-I. It obeys the hardening non-Gaussian distribution under lower structural damping and the Gaussian distribution under higher structural damping. When the structural damping increases gradually, the PDFs of the response

gradually change from hardening non-Gaussian distribution to Gaussian distribution. However, the change of acrosswind direction is slower than that of alongwind direction, i.e., $U/f_s B = 8.95$ and $I_u = 6.3\%$. When the acrosswind direction is in non-Gaussian distribution, the alongwind direction has become Gaussian distribution. It may be that the weakening of the coupling effect is faster than that of the VIV. When there is a partial self-excited force in the acrosswind direction, the coupling effect almost disappears, and the alongwind direction becomes a random vibration.

3.4. Correlation between alongwind and acrosswind responses

Fig. 15 illustrates the phase plane trajectories and correlation coefficients at different reduced wind speeds for uniform and turbulent flow, with structural damping ratios set at 0.09%. The evidence to reveal whether the alongwind response is coupled with the acrosswind response is provided by these figures.

Before VIV-I, there was a weak correlation between the alongwind

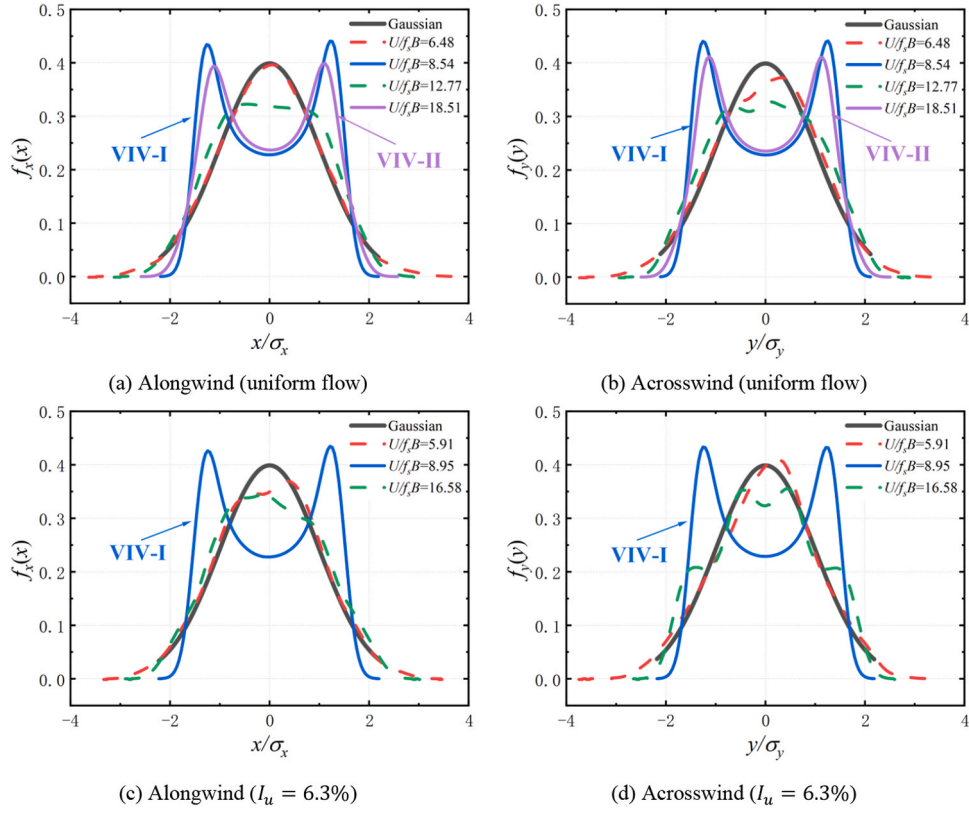


Fig. 13. PDFs under different flow fields and different reduced wind speeds ($\zeta_s = 0.09\%$).

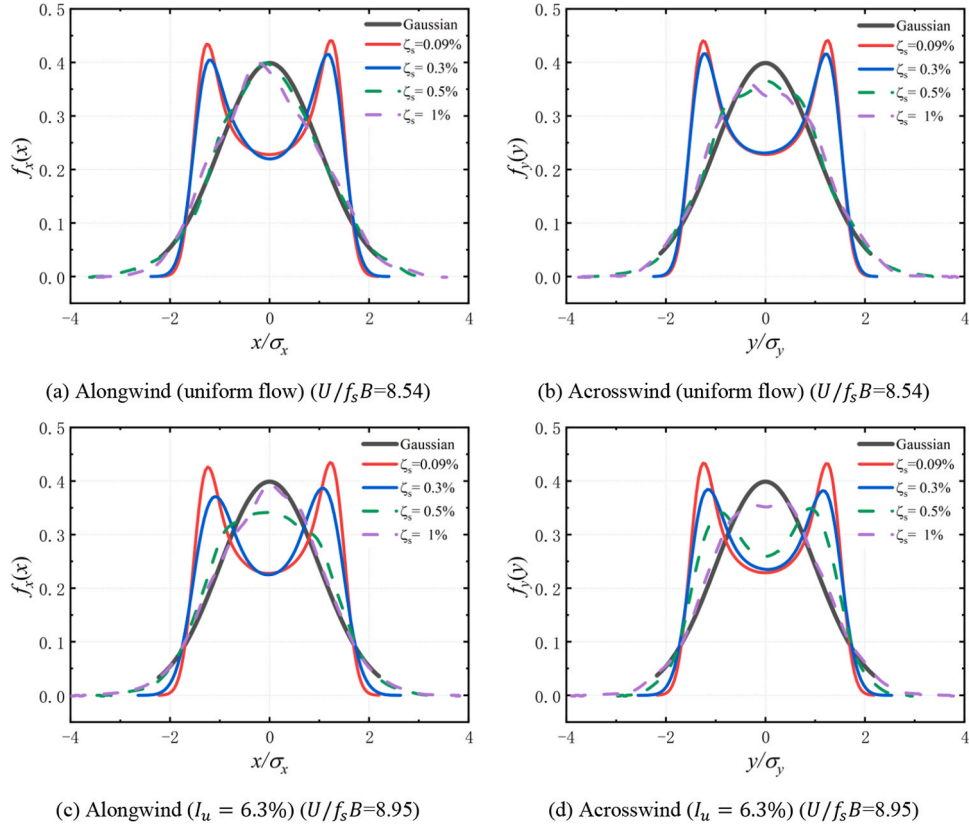


Fig. 14. PDFs of different structural damping in the VIV-I.

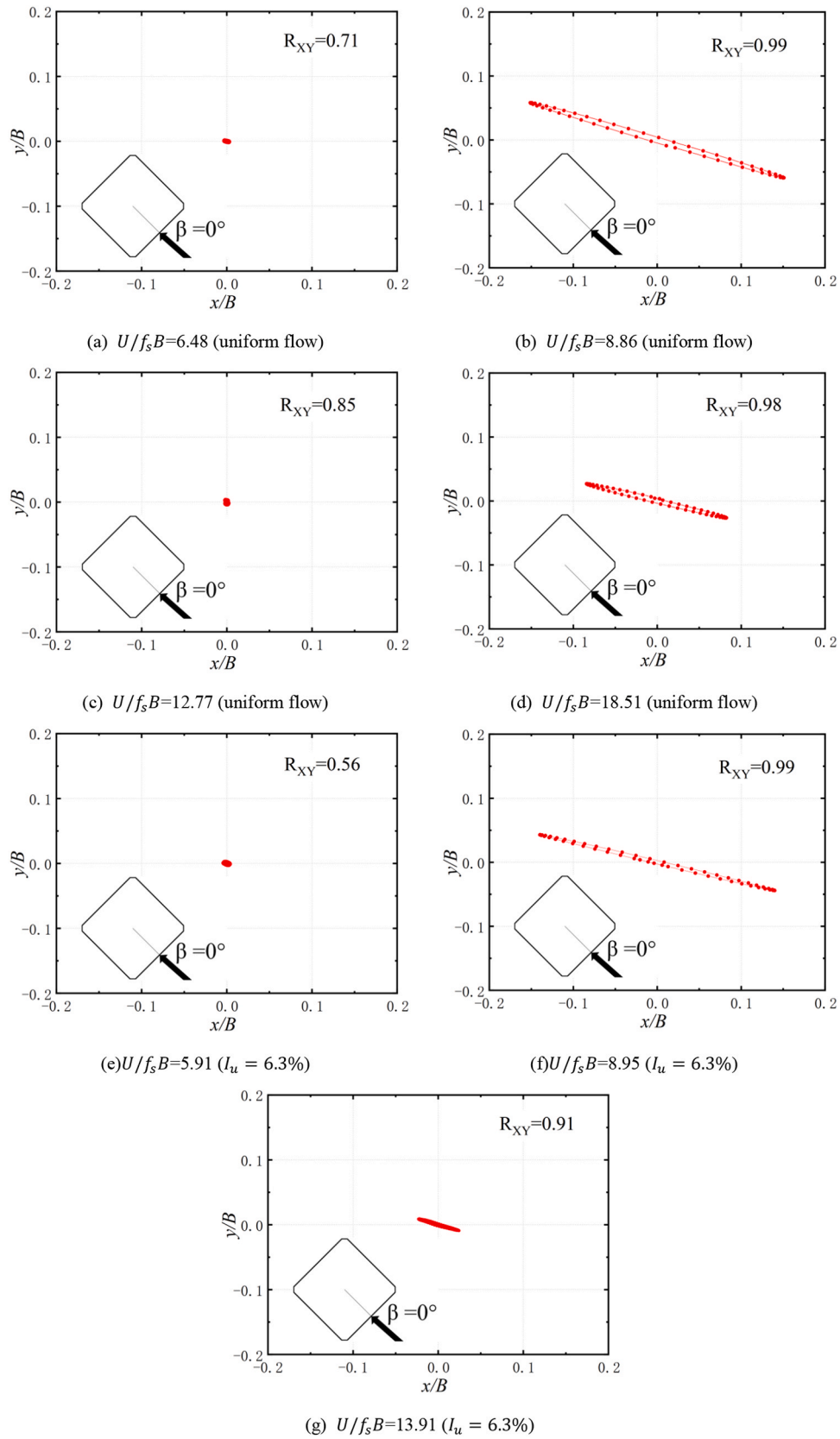


Fig. 15. Phase plane trajectories under different flow fields ($\zeta_s = 0.09\%$).

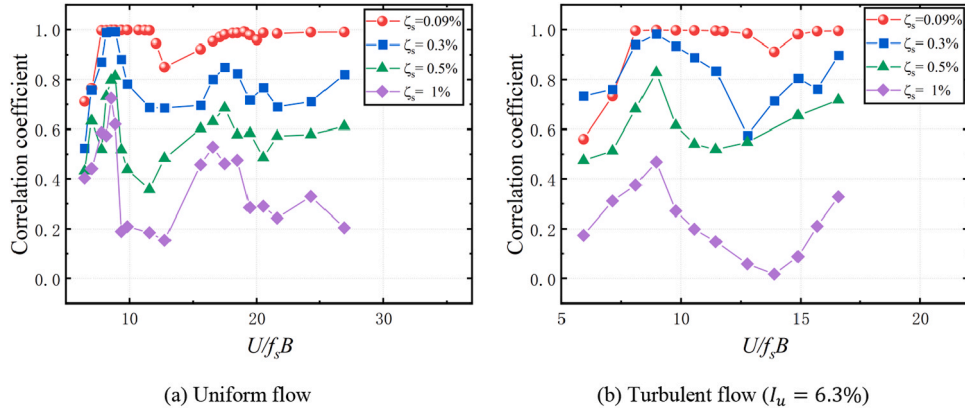


Fig. 16. Correlation coefficients under different flow fields.

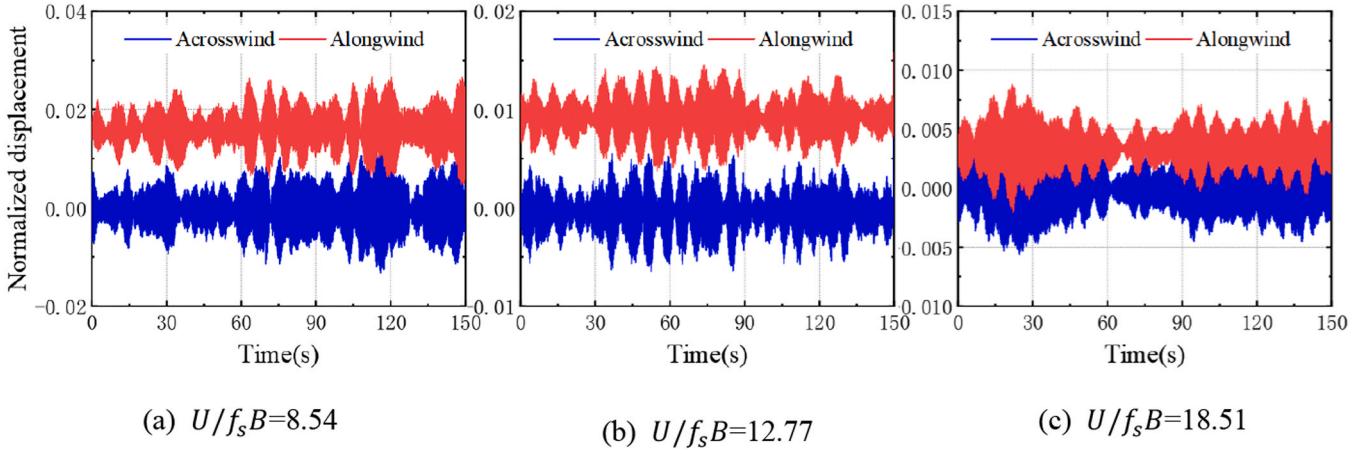
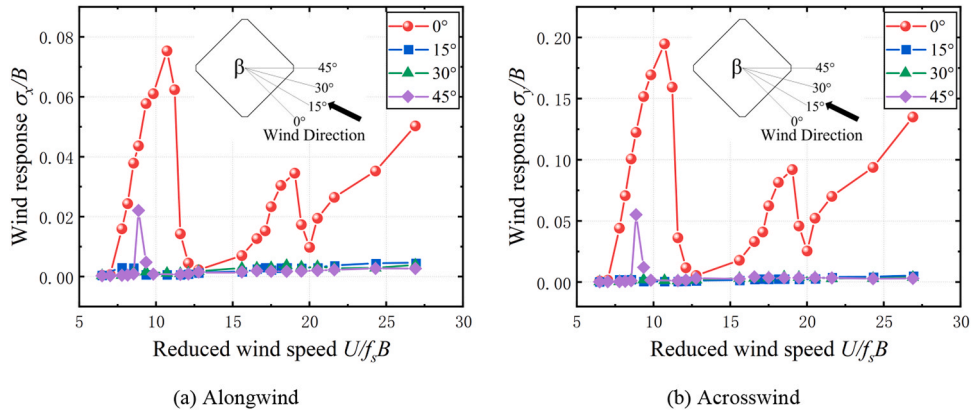


Fig. 17. Displacement response of uniform flow with 30° wind direction.

and acrosswind responses regarding uniform and turbulent flow. The amplitude of alongwind and acrosswind responses is small, and there is no dominant vibration direction in the phase plane trajectory. In VIV-I and VIV-II, the phase plane trajectory is transformed into a narrow ellipse, the amplitude is large, and the phase difference is basically equal to 180°. The corresponding correlation coefficient is almost equal to 1, indicating a strong correlation between the two directions. The amplitude of VIV-I in uniform flow is more significant than that of VIV-II. Between VIV-I and VIV-II, the correlation between the alongwind and acrosswind responses is moderate under the uniform flow, and the correlation coefficient value is between the two extremes. In the tur-

bulent field, after VIV-I, the correlation is weakened.

As shown in Fig. 16(a), in uniform flow, when $\zeta_s = 0.09\%$, the correlation coefficient is close to 1 at almost all wind speeds. For other structural damping ratios, the correlation coefficient significantly increases at vortex-resonance wind speeds compared to different wind speeds. As the structural damping ratio increases, the correlation coefficient notably decreases. From Fig. 16(b), in turbulent flow, the correlation coefficient increases when the wind speed is close to the VIV region, and decreases with the wind speed away from the VIV region. Similarly, as the structural damping increases, the correlation is further weakened. The trend of the correlation coefficient corresponds to the

Fig. 18. STDs of displacement response under different wind directions of uniform flow ($\zeta_s = 0.09\%$).

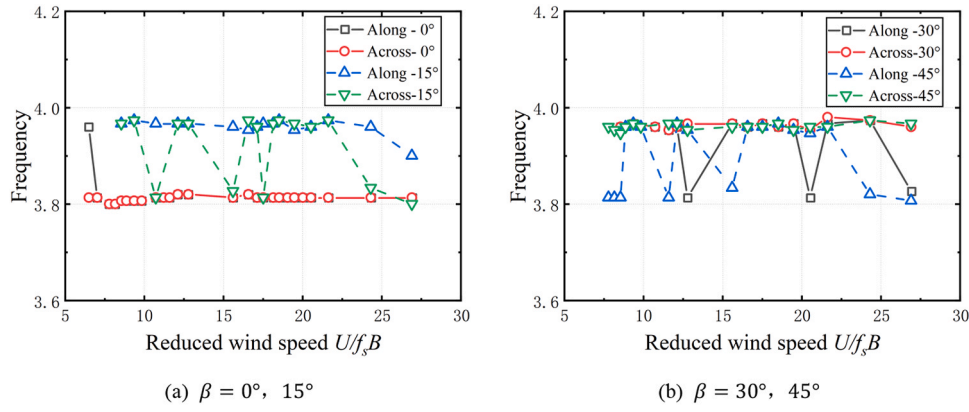


Fig. 19. Dominant vibration frequency of response under different wind directions ($\zeta_s = 0.09\%$).

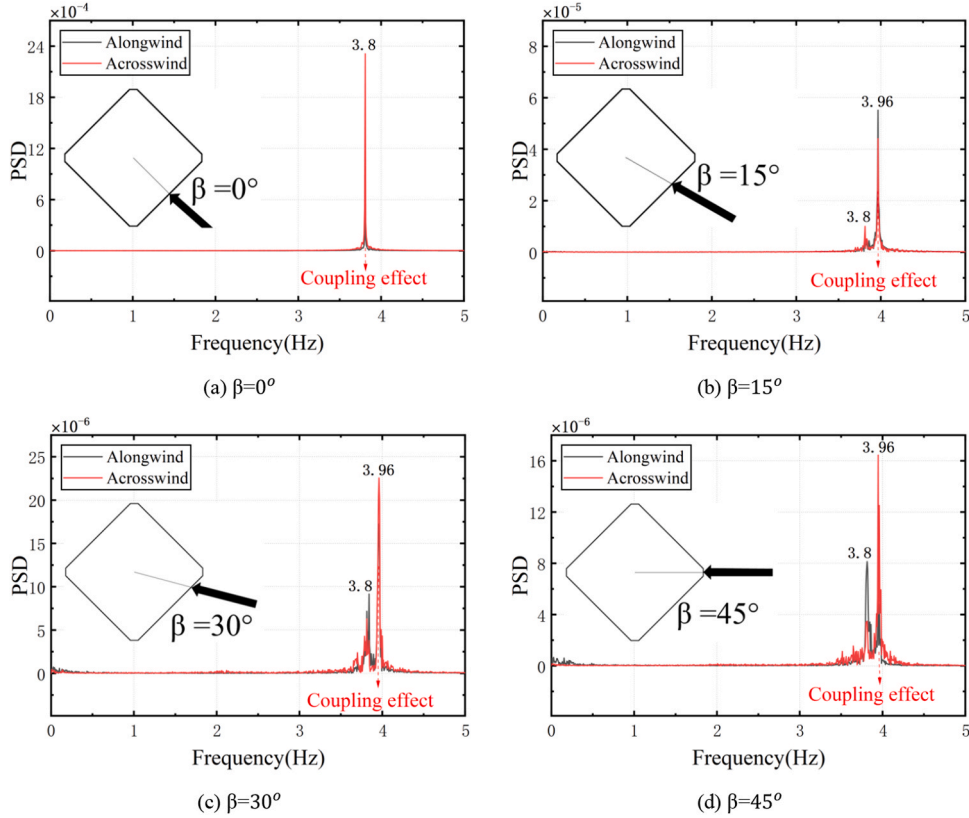


Fig. 20. PSD of different wind directions in the VIV-I ($\zeta_s = 0.09\%$, $U/f_s B = 8.54$).

change of STD. In the VIV region, STD gradually increases, and the correlation coefficient gradually increases. In the non-VIV region, STD gradually decreases, and the correlation coefficient decreases. The coupling effect is strong in the galloping region, and the correlation coefficient becomes larger.

4. The effect of wind direction on coupled vibration

Fig. 17 shows the normalized acrosswind displacement y/B and alongwind displacement x/B under three different reduced wind speeds at uniform flow when the structural damping is 0.09% and wind direction is 30° . The displacement response of VIV and non-VIV regions is dominated by random response, which is different from the result when wind direction is 0° .

Fig. 18 shows the displacement of STDs $\sigma_x/B, \sigma_y/B$ at different wind

directions with 0.09% structural damping under uniform flow. The results show that at the same reduced wind speed, the displacement response of the other three angles is significantly reduced compared with the 0° wind direction. When the wind direction is 15° or 30° , the VIV is not observed, and the response is random response. At 45° wind direction, there is a very narrow peak in VIV-I. For the variable cross-section bridge tower, the wind direction angle will significantly affect the VIV response.

Fig. 19 shows the dominant frequency at different wind directions in uniform flow. With the change of wind direction, the difference in dominant frequency becomes larger, indicating the weakening of the coupling effect. The dominant frequency of the coupling vibration at 0° wind direction is close to the natural frequency of the acrosswind direction. In other wind directions, the dominant frequency is close to the alongwind wind direction. This may be because the variable cross-

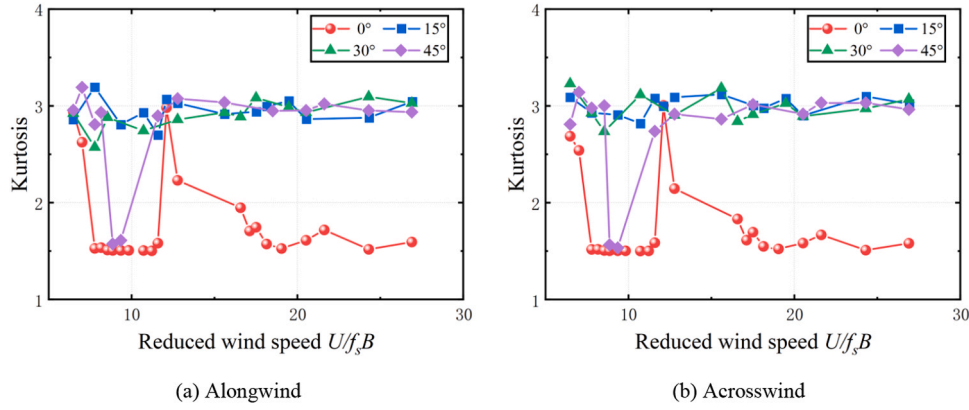


Fig. 21. Kurtosis of uniform flow under different wind directions ($\zeta_s = 0.09\%$).

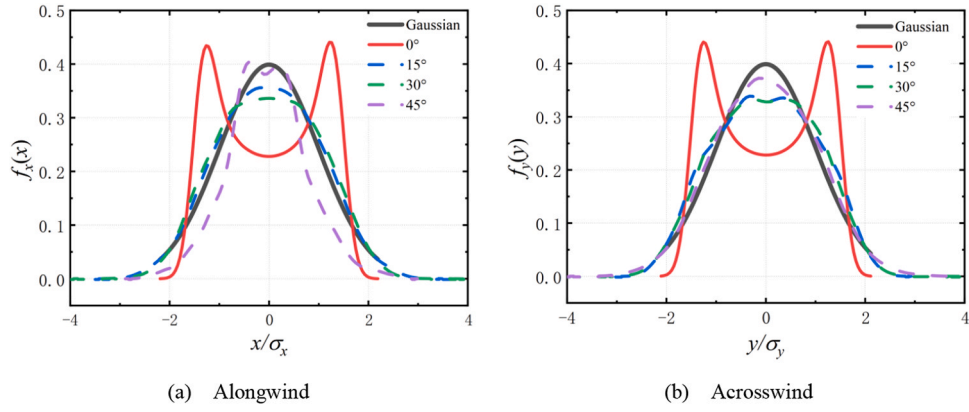


Fig. 22. PDFs of different wind directions in the VIV-I ($\zeta_s = 0.09\%$, $U/f_s B = 8.54$).

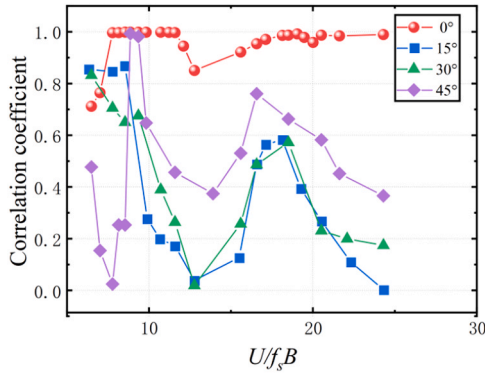


Fig. 23. Correlation coefficient of uniform flow under different wind directions.

section bridge tower for different incoming wind directions will produce different vortex shedding and flow around, thus affecting the vortex vibration and coupling effect.

The coupling effect between the alongwind and acrosswind under different wind directions is further investigated in Fig. 20. At the wind direction of 15° , 30° and 45° , the coupling effect is weak, and the energy at the alongwind and acrosswind natural frequencies and low-frequency parts is large, resulting in the response dominated by random vibration. Different from 0° , the alongwind and acrosswind energy is mainly concentrated at the alongwind natural frequency. The coupling effect under other wind directions may be mainly determined by the alongwind direction.

Fig. 21 shows the kurtosis under different wind directions under 0.09% structural damping of uniform flow. At 15° , 30° and 45° , the response kurtosis of alongwind and acrosswind is around 3 in VIV and non-VIV regions, indicating the random vibration is dominant. In particular, there are two kurtosis values abruptly changed to 1.5 at 45° degrees wind direction, corresponding to the peak of STD at 45° wind direction.

Fig. 22 shows the PDFs of the displacement response under different wind directions at the same reduced wind speed in VIV-I. In these three wind directions, the response basically obeys the Gaussian distribution. The PDFs of the displacement are abnormal at 45° wind direction, the alongwind direction has non-Gaussian characteristics, and the acrosswind direction is close to the Gaussian distribution, which may be related to a peak in STD. In Fig. 23, it can be found that only the correlation coefficients for the two Peaks at 45° wind direction is close to 1, and the correlation coefficients in other wind directions are small. It shows that the wind direction has a significant effect on the correlation between alongwind and acrosswind response.

5. Conclusions

In this study, a wind tunnel aeroelastic experiment of a variable cross-section bridge tower model was carried out to investigate the coupled alongwind and acrosswind response characteristics under different flow conditions and structural damping ratios. The results show that there is a coupling effect between the response in the alongwind and the acrosswind direction, and the coupling effect will be influenced by the flow conditions and the structural damping ratio. Concrete bridge tower typically exhibits higher damping ratios, i.e., larger than 0.3% , and thus the actual concrete bridge tower will un-

likely experience VIV under normal conditions. It needs to be emphasized that these findings are applicable to stand-alone steel towers with the same shape, where lower damping ratios are more prevalent. The main findings can be described as follows:

- (1) Two distinct VIV regions can be observed under the condition of low structural damping in uniform flow, and the response of VIV-I (lower $U/f_s B$) is greater than that of VIV-II (higher $U/f_s B$). The coupling resonance occurs in the VIV region, and the alongwind and acrosswind directions are dominated by steady harmonic vibration. Near the VIV region, the vibration is a combination of harmonic and random vibration, and with the increase of turbulence intensity and structural damping, random vibration gradually becomes significant.
- (2) Higher structural damping will weaken the coupling effect of VIV in alongwind and acrosswind directions and also reduce the response of the structure. The increase of the structural damping will also increase the dominant frequency of the coupling resonance. The increase in turbulence intensity will suppress the VIV and reduce the structural response. Still, it will deteriorate the critical wind speed of galloping and cannot eliminate the coupling effect during galloping.
- (3) VIV and coupling effects are obvious at 0° wind direction rather than in other wind directions (e.g., 15° , 30° , and 45°). When $\beta=0^\circ$, the coupling effect is mainly determined by the acrosswind direction. The alongwind and acrosswind energy is concentrated primarily in the acrosswind dominant vibration frequency, and the coupling effect in other wind directions is mainly determined by the alongwind direction. The reason is that the coupling effect transfers energy between acrosswind and acrosswind directions.

CRediT authorship contribution statement

Shaopeng Li: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Peng Wang:** Writing – original draft, Investigation, Formal analysis. **Yanchi Wu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Yunfeng Zou:** Validation, Software, Funding acquisition. **Qingshan Yang:** Supervision, Resources, Project administration, Funding acquisition. **Jingying Tian:** Visualization, Validation, Investigation. **Yanghao Zhou:** Writing – original draft, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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